

SS-RESfM: master results table

all evaluations, 2026-09-01

Post-BA errors on our track sets, single declared environment. Translation is the aligned camera-position error in scene coordinates; rotation in degrees; N_r/N_c is the fraction of ground-truth cameras surviving robust BA and the 5-pixel quality mask. Self-supervised arms and the primary baseline are reported as seed bands (up to five training seeds, 20–24); reference rows are single runs. Contamination rates measured on our tracks.

Notes. RESfM-scratch is the controlled baseline (our uniform protocol: reprojection-based checkpoint selection, margin 10^{-4}). RESfM-faithful restores the authors' two released secondary values (Accuracy-based checkpoint selection, margin 10^{-3}); it is weaker on five of six datasets, so margins against the primary baseline are lower bounds. RESfM-released are the authors' published weights (MegaDepth directly comparable; OOD on our rebuilt tracks). RESfM at 10^{-4} , 30k is the converged low-rate supervised variant (strongest supervised OOD). ESfM* trains and tests on ground-truth-cleaned tracks (clean-track reference, authors' in-place-zeroing protocol). GLOMAP and COLMAP run on the same tracks with known intrinsics; GLOMAP rotations use reflection-aware alignment.

Table 1: Translation error (mean / median) — primary metric. Seed bands: mean $\pm$ std of per-seed dataset means; n seeds in the last block.

Method	MegaDepth (25.4%)	1DSfM (43.3%)	1DSfM-hard (60%)	Strecha (1.7%)	BlendedMVS (3.1%)	Olsson (0.5%)
self-supervised arms (SS-RESfM), seed bands						
madweight	0.50 $\pm$ 0.07/0.15	10.64 $\pm$ 1.55/2.49	18.92 $\pm$ 4.05/13.36	2.65 $\pm$ 0.25/1.16	0.14 $\pm$ 0.02/0.05	3.01 $\pm$ 0.34/1.36
weight	0.51 $\pm$ 0.09/0.20	14.35 $\pm$ 2.97/9.63	22.92 $\pm$ 2.79/15.56	1.97 $\pm$ 0.09/0.04	0.14 $\pm$ 0.03/0.02	2.91 $\pm$ 0.33/1.14
weight+TTT	0.36 $\pm$ 0.08/0.18	15.72 $\pm$ 2.10/10.75	22.24 $\pm$ 3.17/15.29	2.00 $\pm$ 0.05/0.12	0.14 $\pm$ 0.02/0.01	8.03 $\pm$ 0.30/0.56
remove	0.51 $\pm$ 0.08/0.13	9.13 $\pm$ 2.17/1.68	12.42 $\pm$ 2.05/10.83	3.20 $\pm$ 0.11/2.03	0.24 $\pm$ 0.06/0.17	3.16 $\pm$ 0.33/1.50
remove+TTT	0.56 $\pm$ 0.07/0.14	8.93 $\pm$ 1.20/2.03	15.15 $\pm$ 0.98/11.30	3.01 $\pm$ 0.07/1.70	0.22 $\pm$ 0.10/0.18	3.66 $\pm$ 0.35/1.70
supervised baselines and references						
RESfM-scratch (primary)	0.37 $\pm$ 0.12/0.07	10.50 $\pm$ 1.45/3.39	19.16 $\pm$ 3.42/16.45	0.39 $\pm$ 0.72/0.01	0.35 $\pm$ 0.04/0.33	7.37 $\pm$ 2.63/0.95
RESfM-faithful (authors' cfg)	0.50/0.11	9.75/2.81	17.58/12.30	0.14/0.01	0.37/0.29	8.76/1.08
RESfM-released (authors' weights)	0.21/0.09	11.12/3.60	15.29/12.46	0.20/0.01	0.33/0.31	9.10/0.17
RESfM at 1e-4, 30k budget	0.36/0.07	6.12/2.14	13.88/10.63	1.64/0.01	0.18/0.00	8.27/0.11
RESfM-faithful at 1e-4	0.60/0.17	6.72/2.57	16.21/12.74	0.22/0.01	0.17/0.00	8.11/0.12
ESfM* (clean tracks, authors' protocol)	0.46/0.05	13.84/4.29	15.20/16.23	1.16/0.95	0.33/0.27	—
classical						
GLOMAP (global classical)	3.34/3.46	28.21/24.17	35.44/37.62	0.05/0.01	0.34/0.30	3.14/0.09
COLMAP (incremental)	—	—	20.90/—	—	—	—

Table 2: Rotation error (mean, degrees).

Method	MegaDepth (25.4%)	1DSfM (43.3%)	1DSfM-hard (60%)	Strecha (1.7%)	BlendedMVS (3.1%)	Olsson (0.5%)
self-supervised arms (SS-RESfM), seed bands						
madweight	3.6	7.7	27.6	12.8	9.0	14.4
weight	3.3	8.2	27.9	7.6	9.2	13.2
weight+TTT	2.6	10.1	30.6	8.0	8.3	11.6
remove	4.0	6.9	12.3	14.4	16.0	13.8
remove+TTT	4.2	6.9	16.4	13.2	13.2	15.8
supervised baselines and references						
RESfM-scratch (primary)	1.9	9.6	27.3	1.6	31.7	12.6
RESfM-faithful (authors' cfg)	3.6	9.8	28.0	1.4	39.6	14.9
RESfM-released (authors' weights)	1.9	10.6	20.6	2.0	31.9	16.5
RESfM at 1e-4, 30k budget	3.3	7.7	30.2	8.3	14.2	11.8
RESfM-faithful at 1e-4	4.0	7.7	27.3	1.6	10.7	12.2
ESfM* (clean tracks, authors' protocol)	4.4	13.8	17.1	10.7	34.0	—
classical						
GLOMAP (global classical)	98.0	141.0	86.4	0.4	29.4	28.3
COLMAP (incremental)	—	—	—	—	—	—

Table 3: Registered cameras N_r/N_c .

Method	MegaDepth (25.4%)	1DSfM (43.3%)	1DSfM-hard (60%)	Strecha (1.7%)	BlendedMVS (3.1%)	Olsson (0.5%)
self-supervised arms (SS-RESfM), seed bands						
madweight	80%	70%	43%	91%	92%	86%
weight	78%	79%	40%	98%	94%	90%
weight+TTT	76%	81%	41%	99%	94%	90%
remove	72%	58%	29%	97%	88%	79%
remove+TTT	70%	58%	32%	96%	90%	79%
supervised baselines and references						
RESfM-scratch (primary)	84%	77%	34%	99%	92%	89%
RESfM-faithful (authors' cfg)	83%	78%	35%	100%	93%	90%
RESfM-released (authors' weights)	83%	76%	30%	98%	91%	85%
RESfM at 1e-4, 30k budget	81%	72%	32%	100%	95%	94%
RESfM-faithful at 1e-4	82%	72%	36%	98%	62%	92%
ESFM* (clean tracks, authors' protocol)	87%	79%	33%	100%	95%	—
classical						
GLOMAP (global classical)	96%	96%	80%	100%	86%	100%
COLMAP (incremental)	—	—	64%	—	—	—

Table 4: Seeds per cell (evaluation replicates behind each band; 1 = single run / reference row).

Method	MD	1DSfM	hard	Strecha	BMVS	Olsson
self-supervised arms (SS-RESfM), seed bands						
madweight	5	3	5	5	5	5
weight	5	5	5	5	5	5
weight+TTT	3	5	5	5	5	5
remove	5	5	5	5	5	3
remove+TTT	5	5	5	5	5	3
supervised baselines and references						
RESfM-scratch (primary)	5	5	5	5	5	5
RESfM-faithful (authors' cfg)	1	1	1	1	1	1
RESfM-released (authors' weights)	1	1	1	1	1	1
RESfM at 1e-4, 30k budget	1	1	1	1	1	1
RESfM-faithful at 1e-4	1	1	1	1	1	1
ESFM* (clean tracks, authors' protocol)	1	1	1	1	1	0
classical						
GLOMAP (global classical)	36sc	8sc	5sc	4sc	4sc	39sc
COLMAP (incremental)	0	0	5sc	0	0	0